Hole-Side Airgap Integration as Enabler for 3D NAND Flash Z-Pitch Scaling

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Abstract—We demonstrate for the first time a scalable airgap introduced from the memory hole region as a 3D NAND flash z-pitch scaling enabler. The proposed integration scheme successfully forms airgaps with controlled depth extension in between word lines, offering great potential for manufacturability and advancing storage density and performance. We show that the introduction of these airgaps in a 30nm z-pitch 3D NAND flash process flow effectively reduces WL-WL interference and has no observable impact on the programming operation. While single cell erase operation shows degradation with airgaps, the conventional "all cells erase" scheme is much less impacted. Comparable reliability performance is obtained for devices with and without airgaps. Our results suggest that airgaps can be a viable solution for extreme z-pitch scaling in future 3D NAND flash technologies.

Keywords—3D NAND, Flash, z-pitch scaling, Memory, Airgap, Reliability

I. Introduction

Three-dimensional (3D) NAND flash memories have revolutionized the flash memory industry over the last decade, thanks to steadily increasing storage density at lower cost per bit [1]. This is enabled by vertical tier stacking, which increases the number of memory cells in a string. As this trend continues, however, etch and deposition process complexities and costs mount as well [2]. Thus, tier pitch (z-pitch) scaling is pursued to further decrease storage bit cost. However, scaling gate lengths and inter-gate spacings can lead to enhanced cell-to-cell interference (C2C) with increased program charge interference from neighboring cells. This causes threshold voltage (V_T) shifts that hamper optimal cell operation. To address these issues, TCAD modeling has found the introduction of air gaps between the word-lines (WL) to be a promising strategy for aggressive z-pitch scaling [3].

Recently, an airgap (AG) integrated from the slit region has been demonstrated together with a confined nitride storage node [4]. The proposed process scheme consists of fully removing the inter-gate dielectric via the slit region, then recessing the gate stack and subsequently applying a partial oxide fill resulting in a sealed AG between the WLs. This process scheme suffers from serious limitations, namely the storage node confinement is not scalable due the isotropic recess process, and a large variation in shape and position of the AG. When scaling the inter-gate dielectric, the remaining AG oxide would occupy a significant part of the Inter-Gate-Spacing (IGS), limiting AG reproducibility and scalability.

In this work, we propose a new scalable and controllable process scheme for AG formation. The proposed scheme in addition allows for introducing the AG self-aligned to confined storage nodes processed using different methods discussed elsewhere [5]. This enables aggressive tier pitch scalability without compromising the overall string mechanical stability and with good uniformity.

As proof of concept, we realize a full AG introduced from the memory-hole region and placed controllably between the WLs. The processing is demonstrated structurally and electrically. We first show a detailed description of the process flow of the test vehicle. Then, we report on the electrical performance of the fabricated devices. We use TCAD simulations to provide a good understanding of the electrical behavior of our proposed novel AG integration scheme and its impact on the memory device operation.

II. Test Vehicle

In this study we use a three-layer 3D test vehicle that follows the BiCs-like structure fabrication approach.

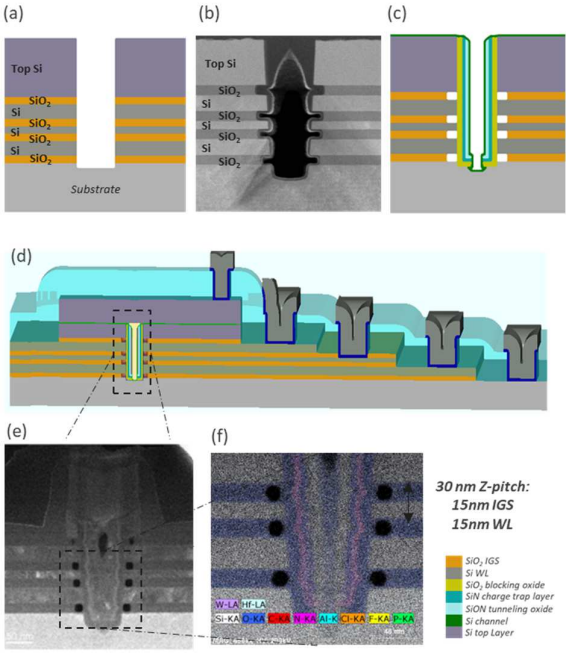


Fig. 1. 3D Integration process flow of AG: (a) Mold stack deposition and memory-hole etch, (b) Controlled recess of the IGS oxide from the memory-hole region, (c) AG formation, gate stack deposition and bottom opening, and channel formation, (g) Final 3D structure after drain, staircase and contact formation with indication of the AGs position, (e) transmission electron microscopy (TEM) and Energy-dispersive X-ray spectroscopy (EDS) images of the formed airgaps.

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Our process flow starts with depositing 15 nm oxide inter-gate spacing (IGS) and poly Si word-line layer stack (OPOP). The electrical performances of the center gate (CG) are evaluated while the top-select, and bottom-select transistors serve to control the source-drain junctions allowing for a precise evaluation of the CG. The z-pitch of our CG is 30 nm. Next, cylindrical memory holes are formed through the OPOP mold-stack. From the memory-hole region we controllably recess around 20 nm of SiO_2 IGS as shown in Fig. 1(b) and 1(c). We next form a sealed airgap at those recessed cavities. The ONO gate stack (~10 nm blocking oxide, ~6 nm SiN, ~5 nm tunnel oxide) is deposited along the sidewall of the holes. These depositions do not interfere with the previously formed sealed AGs ensuring the integrity of the latter. An extra deposition of an amorphous Si (aSi) protection layer is used to protect the gate stack on the sidewalls during the bottom opening of the ONO stack. The protection layer is then removed using a controlled dry isotropic etch and selective to the underlaying materials. A 15 nm aSi channel is subsequently deposited and crystallized. The channel is partially oxidized using in situ steam generation (ISSG) to form a thin channel with large grain poly-Si. The drain is then deposited and patterned followed by staircase and contacts formation.

Fig.1(d) shows schematically the end results of the overall integration flow with the novel AG approach. Finally, the transmission electron microscopy (TEM) and Energy-dispersive X-ray spectroscopy (EDS) images of the formed AGs are shown in Fig.1(e) and 1(f) respectively. We confirm from these images that this airgap processing method enables the controlled formation of AGs with no residuals and exclusively in the targeted regions. This approach is hence scalable and can ensure further tight z-pitch scaling.

III. Electrical Performance

Fig. 2 shows the schematic of a typical 3D NAND string measured in this work. Depending on the mode in which the string is operated, a program / erase / read voltage ($V_{applied}$) is applied to the selected cell while an appropriate pass voltage (V_{pass}) is applied to the unselected cells. The V_{pass} must be sufficiently high to ensure complete turn-on of the string.

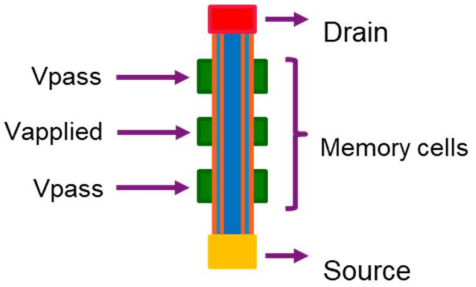


Fig. 2. Schematic of a 3D NAND string. During the read and programming operations, sufficiently high V_{pass} is needed to ensure the channel under unselected cells is turned on.

Fig. 3(a) compares the typical transfer characteristics for 3D NAND cells with and without an AG in the IGS. The CG device described in the previous section is measured in the fresh state at a 1 V drain voltage and V_{pass} of 4 V. A stable AG process flow is evident from the comparable spread in threshold voltage, V_T ($V_{applied}$ at 1 nA drain current) and very similar subthreshold swing (STS) (see Fig. 3(b) and 3(c)). Devices with airgap show slightly lower I_{ON} drive current compared to those without AG as indicated in Fig. 3(c). We expect that the influence of the side gates on the channel under the center gate reduces with the introduction of airgap.

In Fig. 4, we show that the V_{pass} applied to the unselected gates causes a significant V_T shift on the CG, whereas the introduction of an AG greatly reduces this effect. Fig. 4(a) shows the transfer curves at different V_{pass} for devices without an AG, exhibiting a V_T shift analogous to a short channel effect (SCE). In Fig.4(b), on the other hand, the characteristics of the devices with an AG are significantly less affected by the applied V_{pass}. Such insensitivity to neighboring cell interference is crucial for future z-pitch scaling. This resilience to V_{pass} interference is also evident from the distribution plots of CG's V_T as shown in Fig. 5. As V_{pass} increases, the CG V_T reduces, but the impact is significantly lower with the presence of an AG, showing a saturation of V_T at higher V_{Pass}.

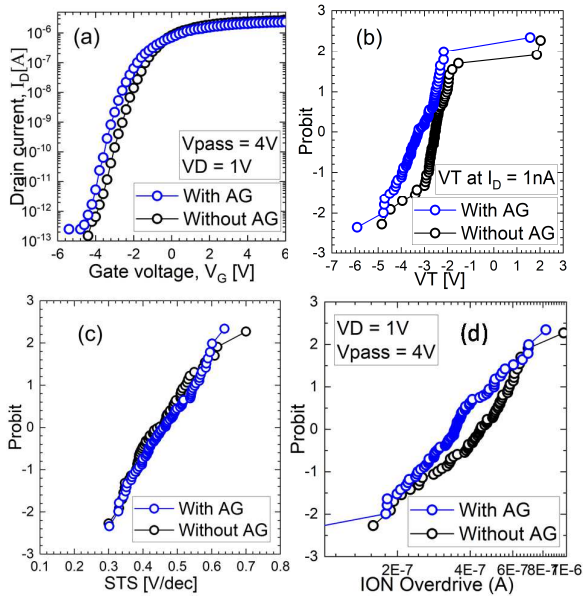


Fig. 3. (a) Typical I_D-V_G curves for devices with airgap compared to those without AG in the IGS. (b) Distribution of threshold voltage, V_T defined as gate voltage at 1nA drain current. (c) Subthreshold swing (STS) distribution. A stable AG flow is achieved (d)Distribution of I_{ON} drive current. Data for devices with 80nm memory hole diameter is shown.

Next, we show the impact of the AG on the memory performance of the CG cell in Fig. 6, by applying an Incremental Step Pulse Programming and Erase (ISPP and ISPE) operation. The presence of the AG has little discernible effect on the program behaviour. However a significant negative effect on the erase saturation is observed with AG. Furthermore, we notice this to be consistent across different memory hole diameters as shown in Fig. 6(c).

To understand these trends, we performed TCAD simulations based on a memory operation model co-developed with Global TCAD Solutions [6, 7]. Fig.7(a-b) shows the simulated impact on the memory operation as the AG is extended from between the gates up to the blocking

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oxide interface in steps of 2 nm. As in the experimental measurements, the ISPP curves are only slightly affected, whereas ISPE from fresh shows a strong degradation of the saturation level. When the AG is extended to the blocking oxide interface (AGExt=4 nm), which is closest to the experimental condition, there is almost no erase occurring, consistent with the measurements in Fig.6(b).

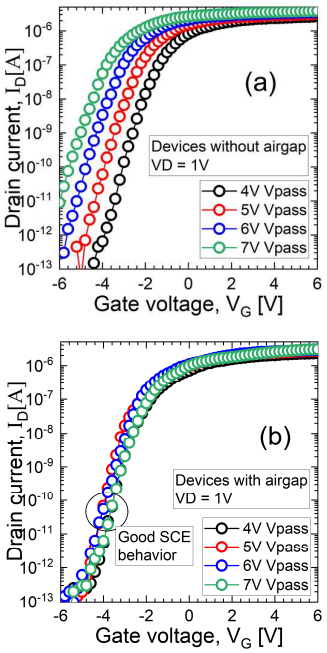


Fig. 4. Transfer curves of CG devices at different V_{pass} on side gates for the case of (a) without airgap, (b) with airgap. Median characteristics from a population of 60 – 70 devices are shown. Data for devices with 80nm memory hole diameter is shown.

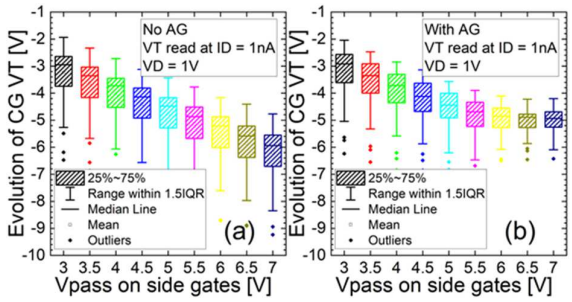


Fig. 5. Box plot showing the change V_T of CG devices at different V_{pass} for the case of (a) without airgap, (b) with airgap. Data for devices with 80nm memory hole diameter is shown.

Fig. 7(c) shows that this is because the presence of the low-k AG enhances the electric field at the WL corners. This increases exponentially the tunnelling probability for unwanted injection of electrons from the WL. These electrons compete with the holes injected from the channel, causing early erase saturation.

Up to this point, we considered ISPE on the CG memory cell with the side gates at constant V_{pass}. However, in actual 3D NAND flash memory arrays a block erase scheme is performed to erase all the memory cells in a given set of strings at the same time. Fig. 8 shows the result applying ISPE at the same time on all the 3 gates in our 3D NAND test vehicle. The erase saturation is still degraded with an AG, but the difference between devices with and without AG reduces from ~5 V to ~2 V after an "all cells erase" scheme. This can be attributed to the cooperation of all gates on the string, benefiting the hole injection from the channel. This shows that single cell erase presents a worst-case scenario and supports the perspective of using an inter-WL AG to alleviate negative effects of z-pitch scaling on cell V_T, with limited detriment to memory window.

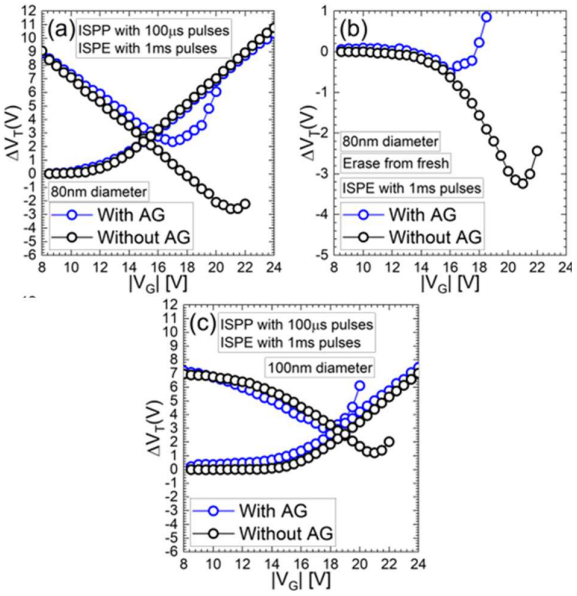


Fig. 6. Memory performance of devices with and without AG.(a, c) Program and erase from program. (b) Erase from fresh. The impact of AG on erase is consistent across different memory hole dimensions. The ISPP and ISPE pulses were applied to the CG with appropriate V_{pass} on side gates.

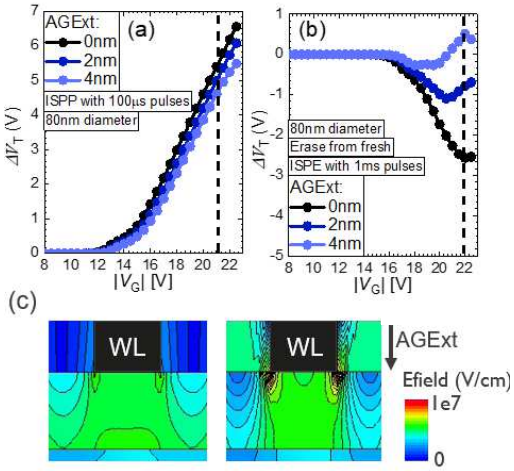


Fig. 7. Simulated AG impact on memory operation for varying AG extension (AGExt) (a) Program, (b) erase from fresh (c) electric field contour plot at $|V_G|$=10 V in no-AG case (left) and for AGExt=4 nm (right) showing increased electric field at the gate corners due to the AG.

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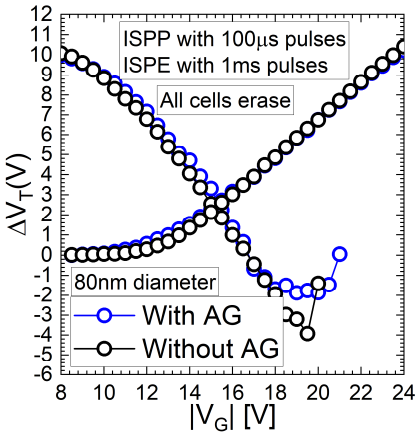


Fig. 8. Program and erase from program using an all cells erase scheme.

Fig. 9 shows the simulated impact of extending the AG beyond the inter-gate space and into the memory stack. The ISPP is projected to degrade as the AG extends into the blocking oxide, due to the reduction of capacitive coupling between the captured charge and the WL, and then saturate with further extension into the CTL. ISPE from fresh on the other hand, is seen to recover for AG extension into the CTL. This indicates that AG extension should be carefully designed and it is optimal when AG is inserted between separated CTLs to meet the required memory window specifications.

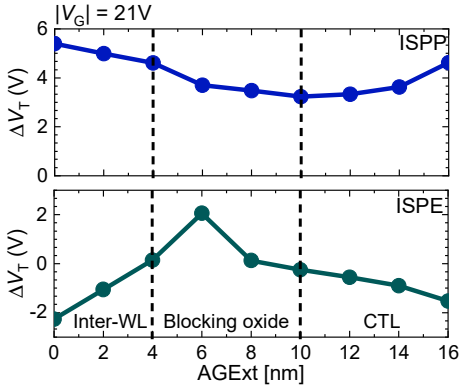


Fig. 9. Simulated impact of AG extension on memory operation, characterized by the achieved DV_T for a fixed program/erase voltage of 21 V (see dashed lines in Fig.7(a-b)). The top AG interface is fixed between the WLs at 4nm from the blocking oxide (see Fig.7(c)). The dashed lines indicate the boundaries between the layers in the memory stack.

Finally, we study the reliability of devices with and without the inter-WL AG in Fig. 10. The measurements show that the presence of AG has no impact on the endurance of the 3D NAND devices up to 1000 cycles. A single cell erase scheme was used in these measurements, which explains the smaller memory window for the AG devices. The charge retention for uncycled devices with AG is in line with that of devices without AG after 85 °C bake (see Fig. 10(b)).

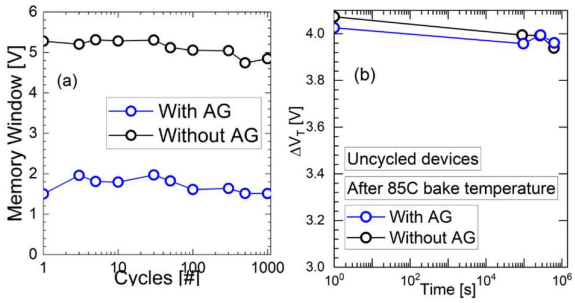


Fig. 10. Comparing devices with and without AG for (a) Memory window evolution as a function of Program/erase cycles, (b) Charge loss after 85 °C bake for uncycled devices, Data for devices with 80 nm memory hole diameter is shown. AG devices show comparable reliability performance.

IV. Conclusion

In this work we have demonstrated for the first time an airgap introduced from the memory-hole as a scaling enabler for future 3D NAND flash memories. This approach is attractive for its non-intrusive and mechanically stable process scheme. Moreover, the key benefit of the proposed scheme is ensuring the accurate placement of the airgap in the desired positions, which makes it compatible with further z-pitch scaling. We found that devices with airgap in the inter-gate-spacing show excellent resistance to WL-WL interference. In addition, the presence of the airgap has no observable impact on programming operation whereas single cell erase is intrinsically degraded. However, by implementing an "all cells erase" scheme, as in the actual product, the negative impact on the erase saturation due to the airgap is reduced. Furthermore, simulations show that AG extension into the region of the CTL mitigates the negative impact on ISPE. Devices with airgap show comparable reliability performance to those without. Thus, when carefully designed, our AG concept can serve as a promising solution to enable aggressive z-pitch scaling of future 3D NAND flash technologies.

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